

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\sanje\anaconda3\lib\site-packages (3.10.6)
```

```
Requirement already satisfied: contourpy>=1.0.1 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (1.3.3)
```

```
Requirement already satisfied: cyclor>=0.10 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
```

```
Requirement already satisfied: fonttools>=4.22.0 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (4.60.1)
```

```
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (1.4.9)
```

```
Requirement already satisfied: numpy>=1.23 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (2.3.5)
```

```
Requirement already satisfied: packaging>=20.0 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (25.0)
```

```
Requirement already satisfied: pillow>=8 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (12.0.0)
```

```
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (3.2.5)
```

```
Requirement already satisfied: python-dateutil>=2.7 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
```

```
Requirement already satisfied: six>=1.5 in c:\users\sanje\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
import matplotlib.pyplot as plt
```

```
x=[1,2,3]
```

```
y=[10,20,30]
```

```
fig, ax = plt.subplots()
```

```
ax.plot([1,2,3], [10,20,30])
```

```
plt.title("Monthly Sales")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales")
```

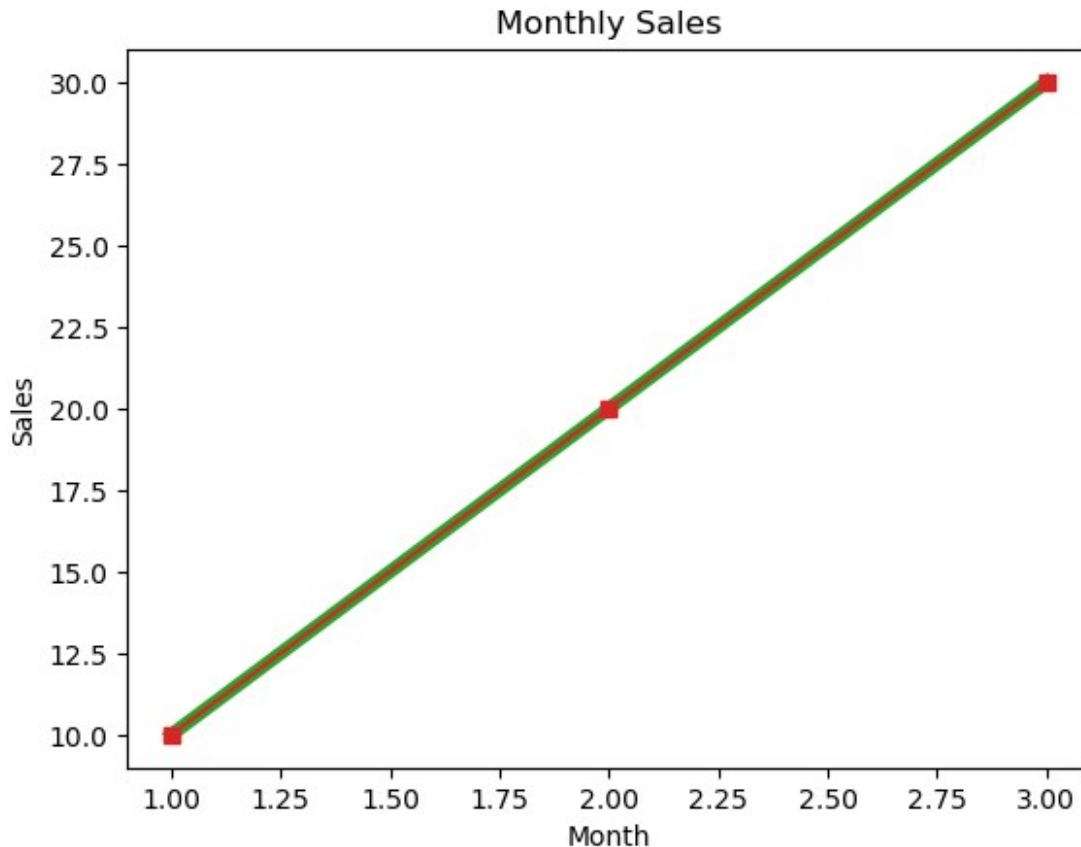
```
plt.plot(x,y, color='green')
```

```
plt.plot(x,y, linestyle='-.')
```

```
plt.plot(x,y, linewidth=5)
```

```
plt.plot(x,y, marker='s')
```

```
plt.show()
```

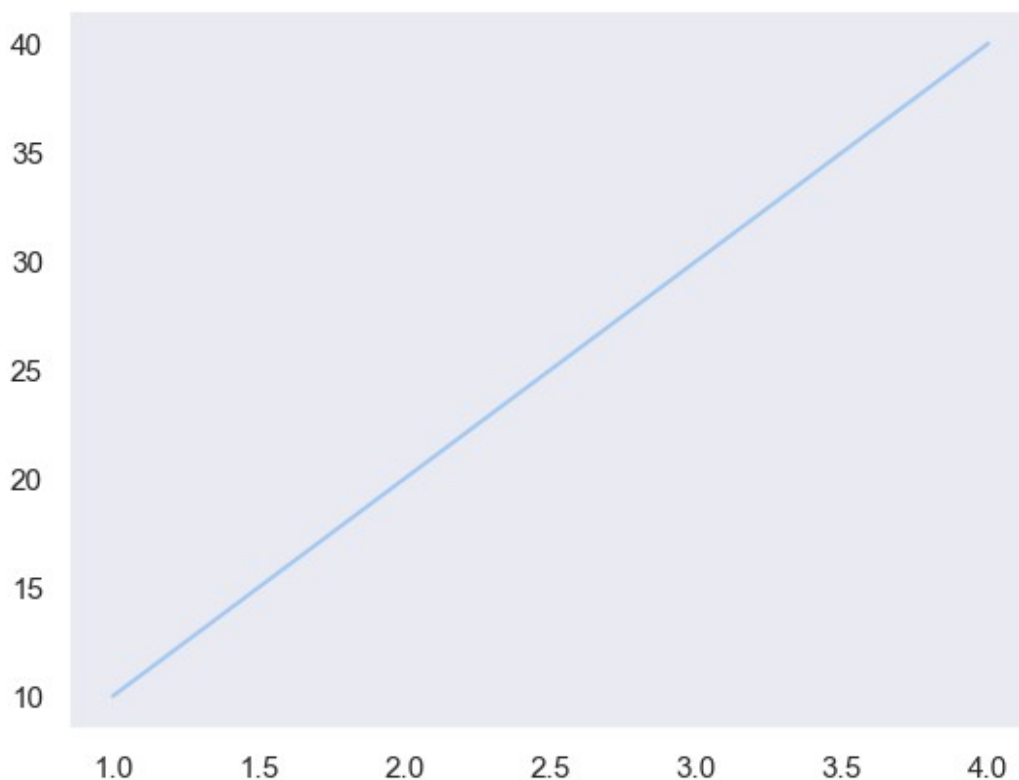


```
!pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\sanje\anaconda3\lib\site-packages (0.13.2)  
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\sanje\anaconda3\lib\site-packages (from seaborn) (2.3.5)  
Requirement already satisfied: pandas>=1.2 in c:\users\sanje\anaconda3\lib\site-packages (from seaborn) (2.3.3)  
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\sanje\anaconda3\lib\site-packages (from seaborn) (3.10.6)  
Requirement already satisfied: contourpy>=1.0.1 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)  
Requirement already satisfied: cycler>=0.10 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)  
Requirement already satisfied: fonttools>=4.22.0 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.60.1)  
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\sanje\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)  
Requirement already satisfied: packaging>=20.0 in c:\users\sanje\
```

```
anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(25.0)
Requirement already satisfied: pillow>=8 in c:\users\sanje\anaconda3\
lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\sanje\
anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\sanje\
anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\sanje\
anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\sanje\
anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\sanje\anaconda3\
lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4-
>seaborn) (1.17.0)
```

```
import seaborn as sns
import matplotlib.pyplot as plt
x = [1,2,3,4,5]
y = [10,20,30,40,50]
sns.set_theme()
sns.set_style("dark")
sns.set_palette("pastel")
sns.lineplot(
    x= [1,2,3,4],
    y= [10,20,30,40]
)
plt.show()
```



```
import seaborn as sns
print(sns.get_dataset_names())

['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes',
'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri',
'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets',
'seaice', 'taxi', 'tips', 'titanic']
```

```
import seaborn as sns
df = sns.load_dataset("titanic")
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked
class \								
0	0	3	male	22.0	1	0	7.2500	S
Third								
1	1	1	female	38.0	1	0	71.2833	C
First								
2	1	3	female	26.0	0	0	7.9250	S
Third								
3	1	1	female	35.0	1	0	53.1000	S
First								
4	0	3	male	35.0	0	0	8.0500	S
Third								

```
who  adult_male  deck  embark_town  alive  alone
```

0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

#Linechart

#what is a line chart?

#a line chart connects data points

```
import matplotlib.pyplot as plt
months=["Jan","Feb","mar","Apr"]
sales=[100,150,200,250]
plt.plot(months,sales,marker="o")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```



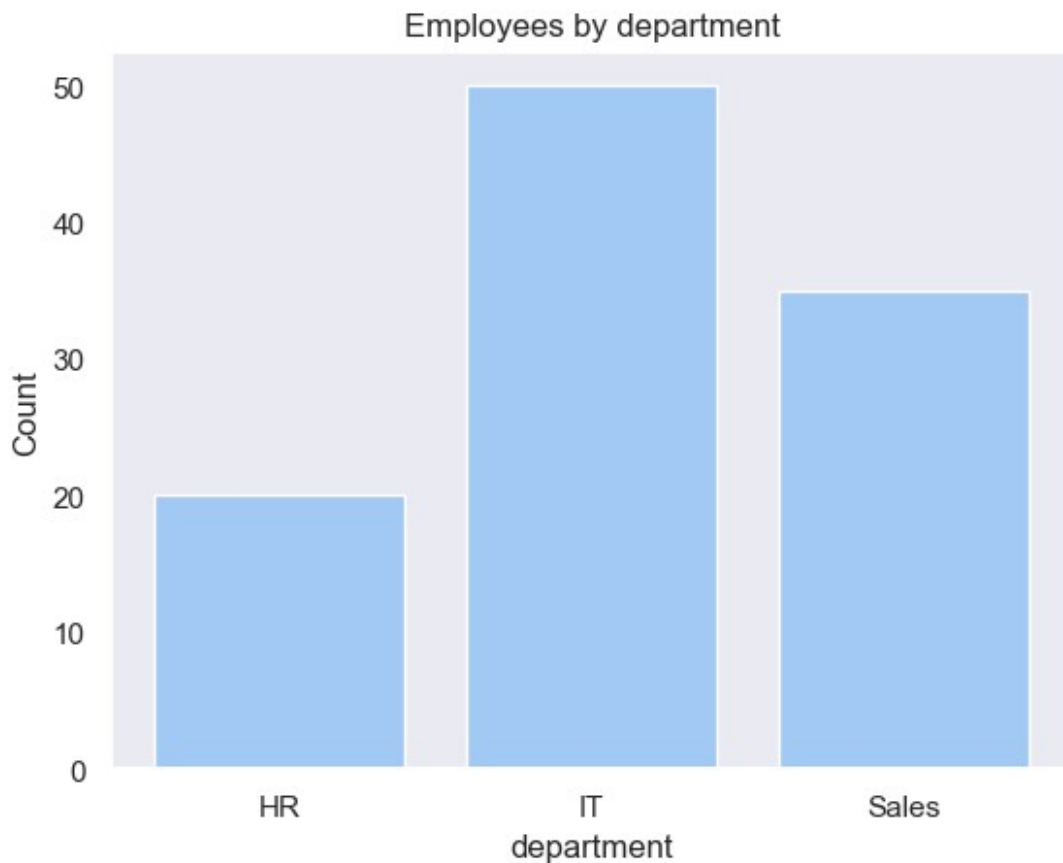
#bar chart

```
import matplotlib.pyplot as plt
```

```

dept=["HR","IT","Sales"]
employees=[20,50,35]
plt.bar(dept,employees)
plt.title("Employees by department")
plt.xlabel("department")
plt.ylabel("Count")
plt.show()

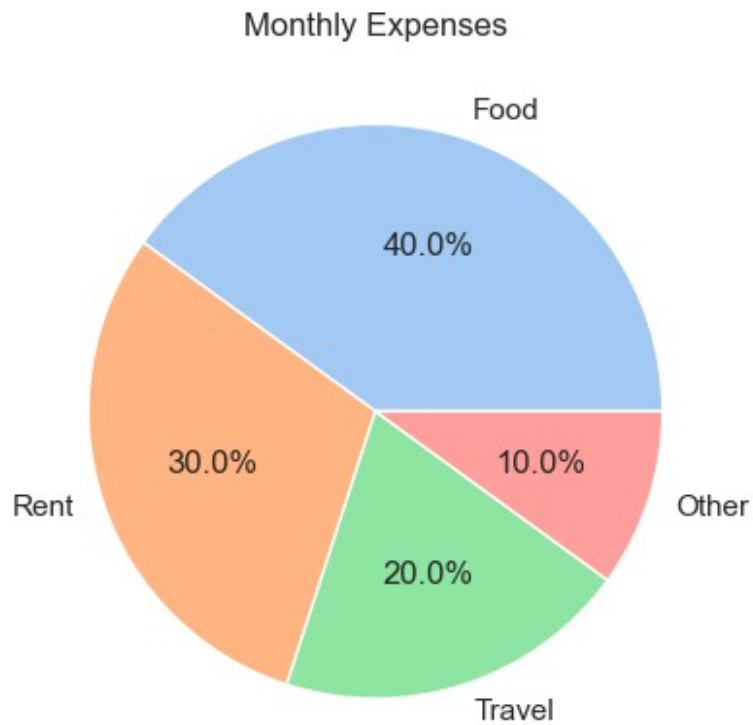
```



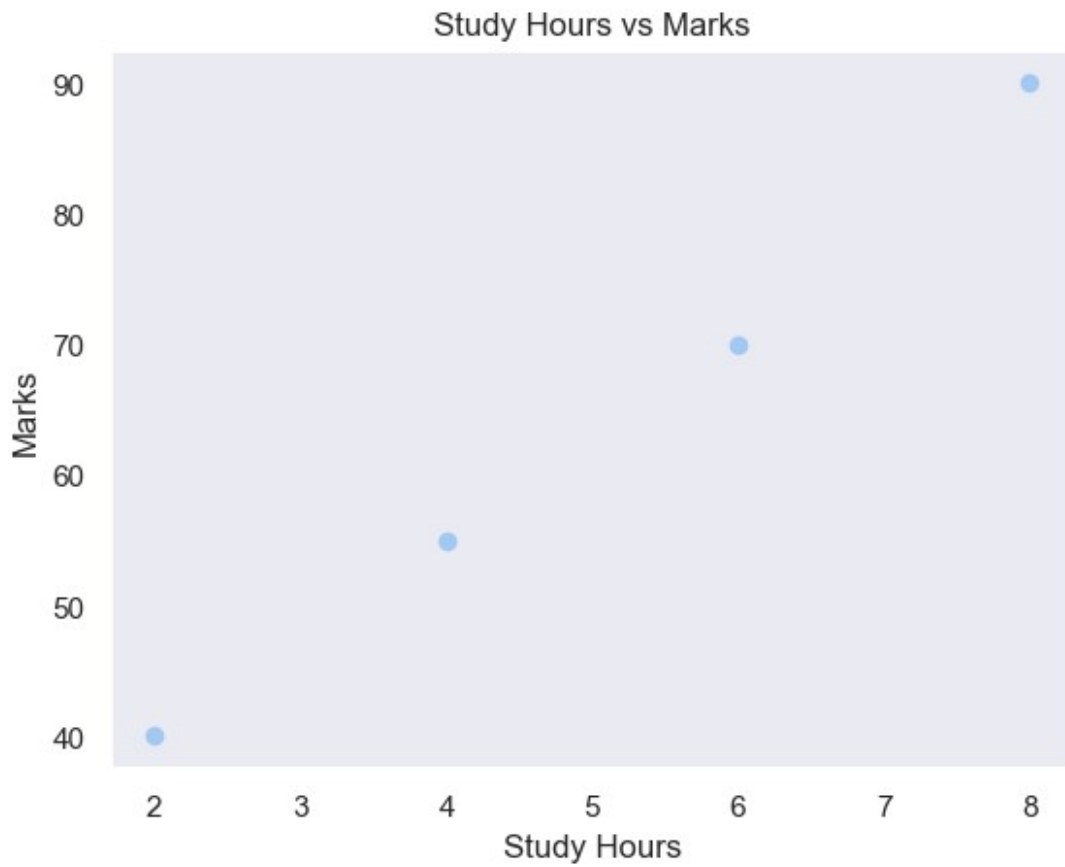
```

import matplotlib.pyplot as plt
labels = ['Food', 'Rent', 'Travel', 'Other']
values = [40, 30, 20, 10]
plt.pie(
    values,
    labels=labels,
    autopct='%1.1f%%'
)
plt.title("Monthly Expenses")
plt.show()

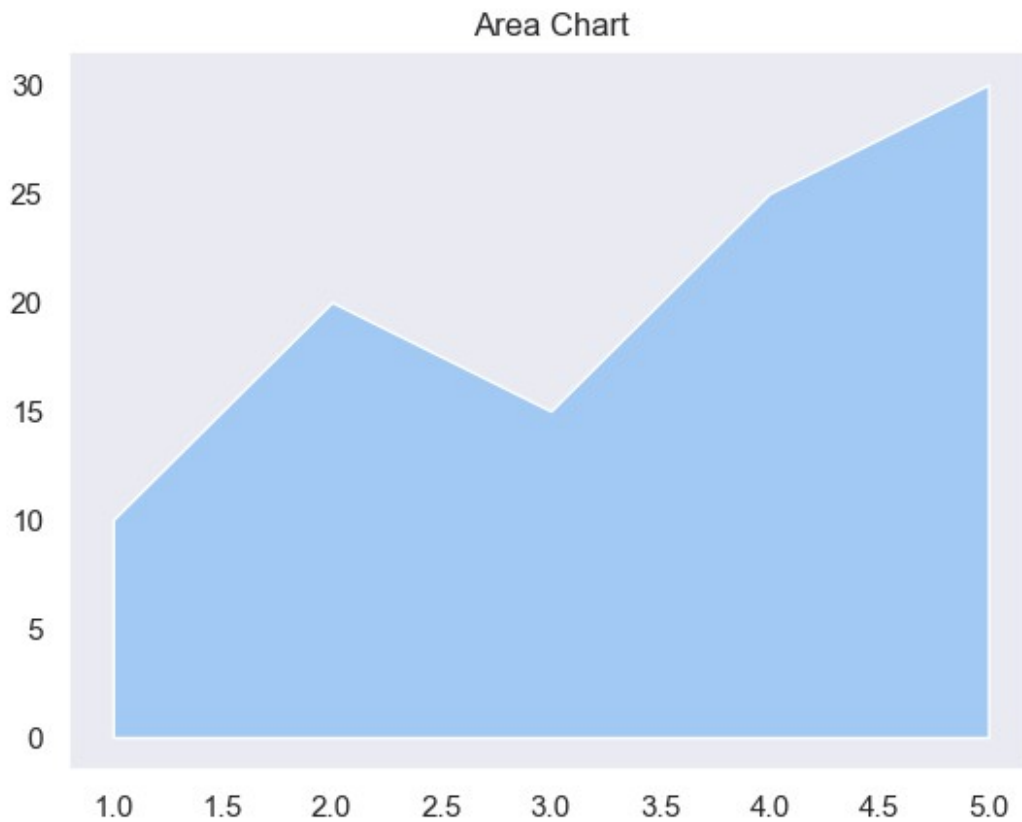
```



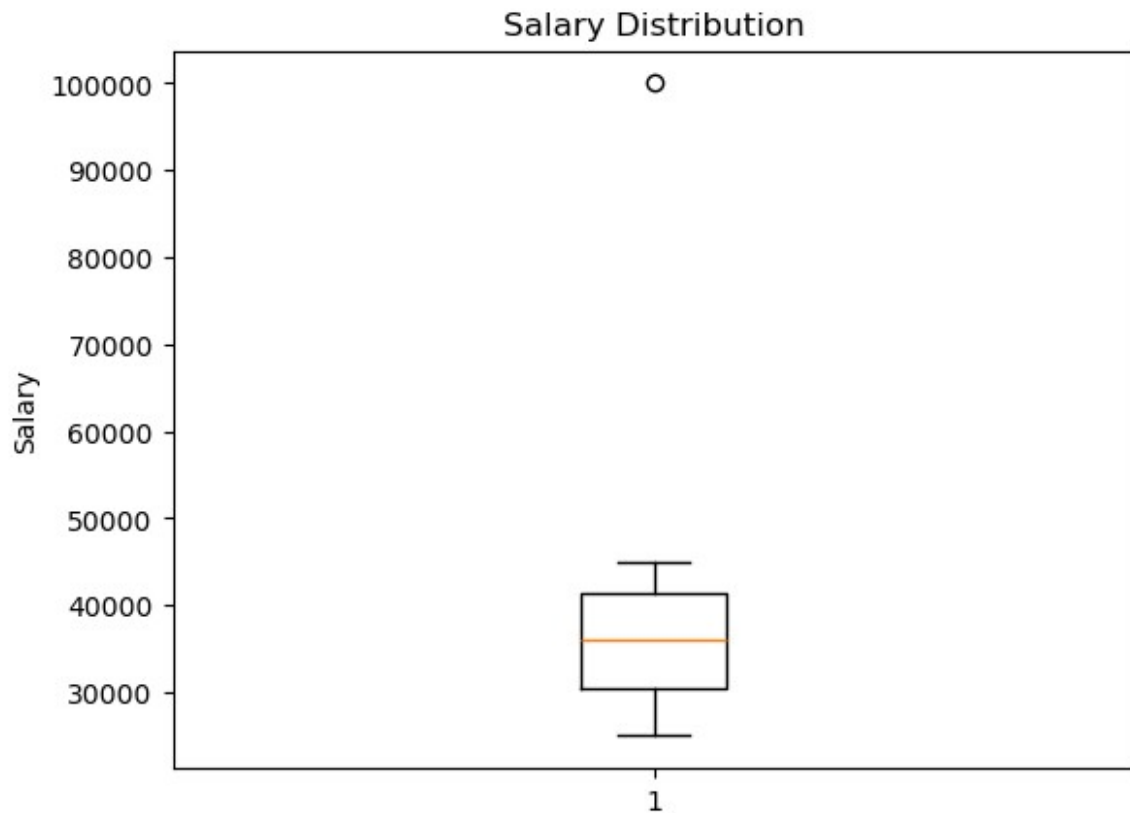
```
import matplotlib.pyplot as plt
hours = [2, 4, 6, 8]
marks = [40, 55, 70, 90]
plt.scatter(hours, marks)
plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.show()
```



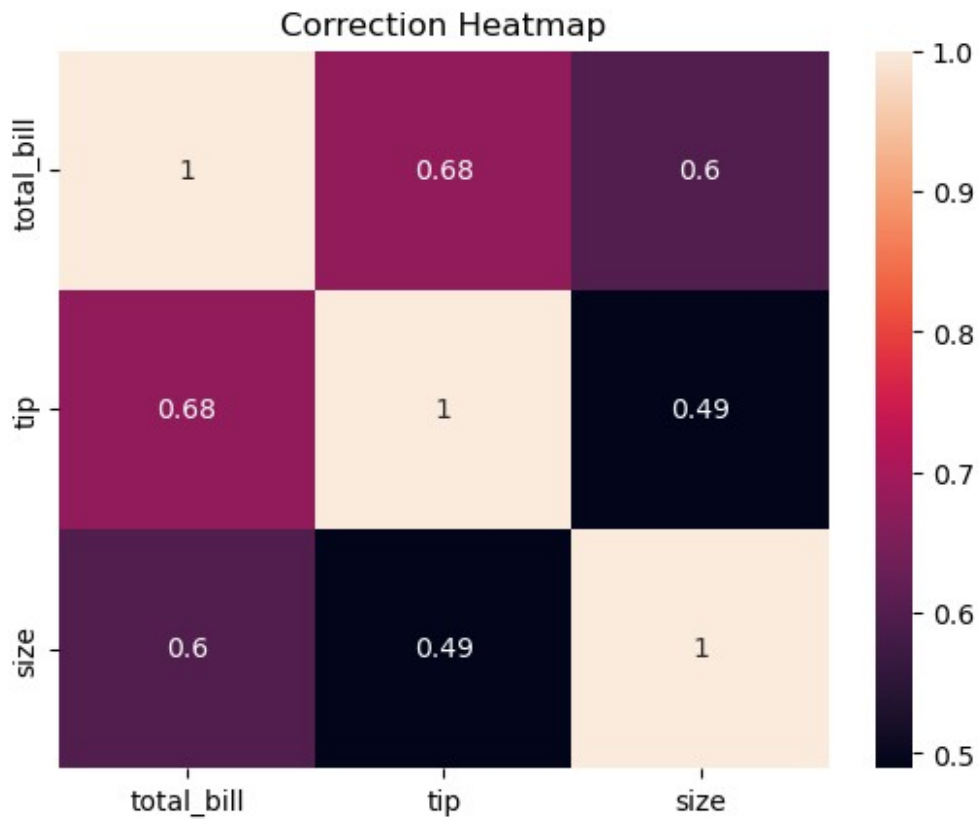
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]
plt.fill_between(x,y)
plt.title("Area Chart")
plt.show()
```

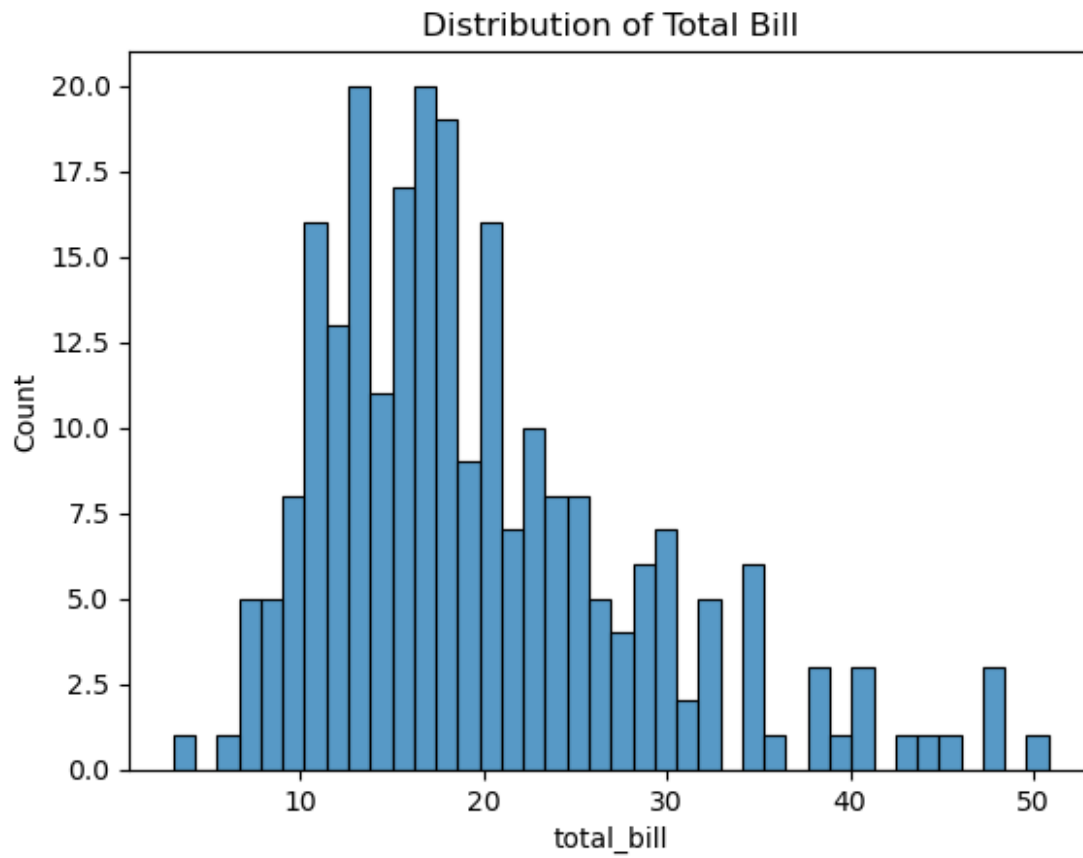
```
import matplotlib.pyplot as plt
salary = [25000, 28000, 30000, 32000, 35000,
          37000, 40000, 42000, 45000, 100000]
plt.boxplot(salary)
plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
```



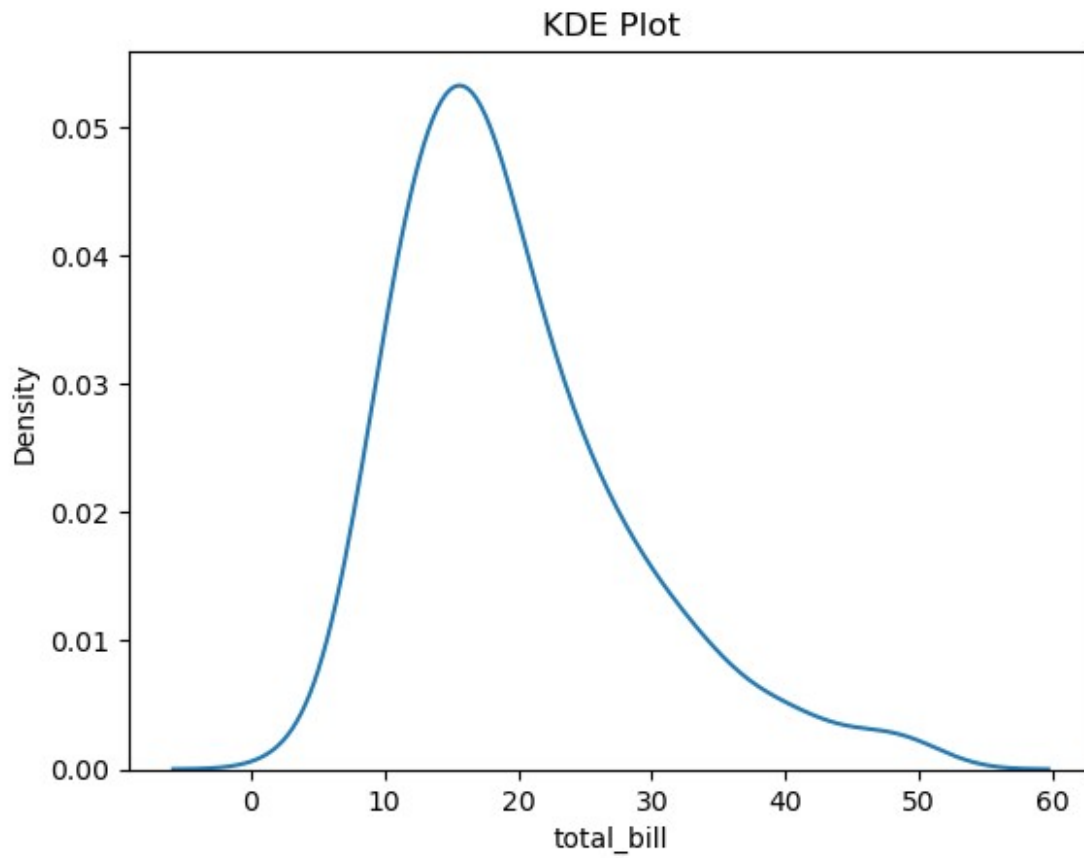
```
import seaborn as sns
import matplotlib.pyplot as plt
tips = sns.load_dataset("tips")
corr = tips.corr(numeric_only=True)
sns.heatmap(corr, annot=True)
plt.title("Correction Heatmap")
plt.show()
```



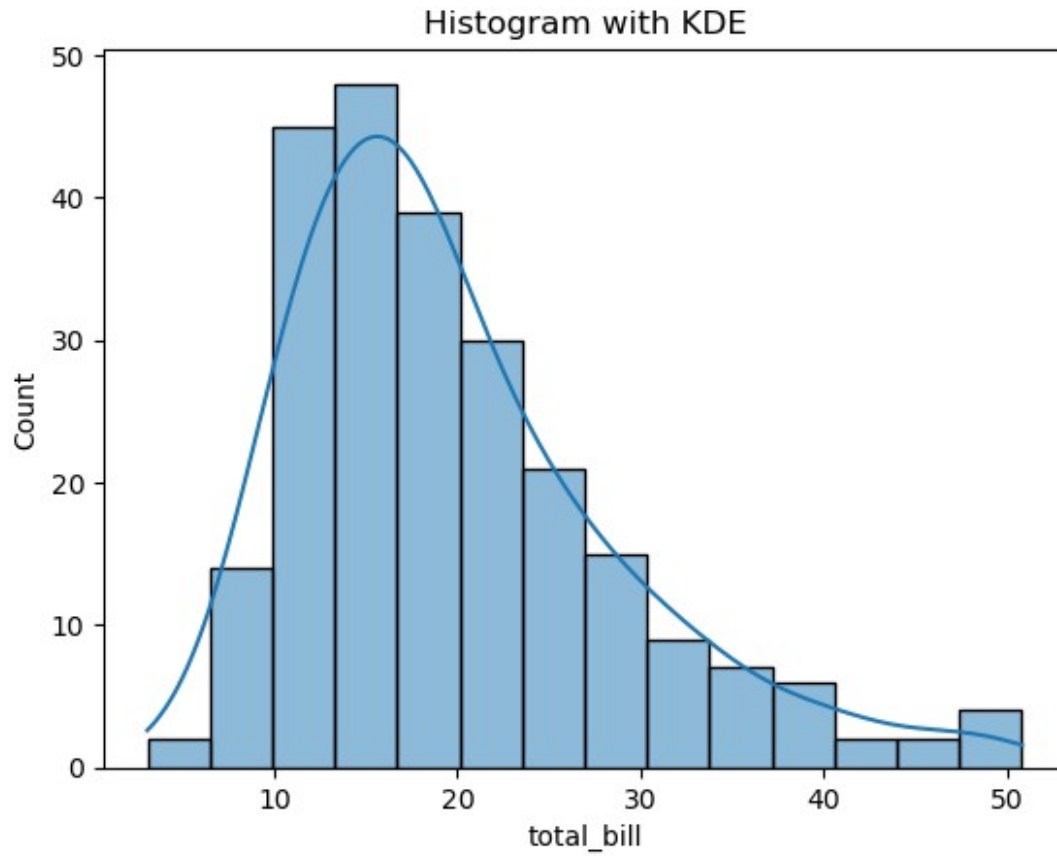
```
sns.histplot(  
    data=tips,  
    x="total_bill",  
    bins=40  
)  
plt.title("Distribution of Total Bill")  
plt.show()
```



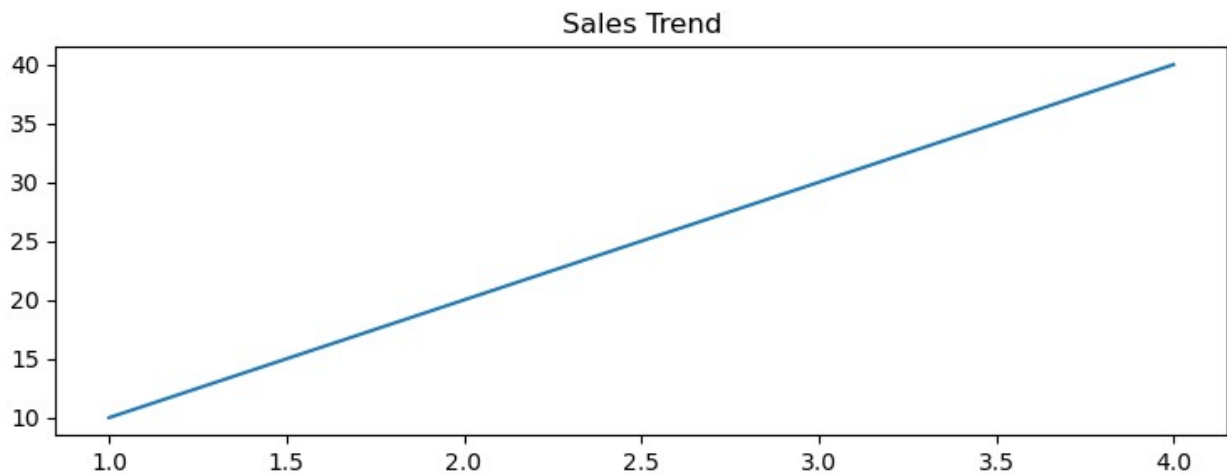
```
import seaborn as sns
import matplotlib.pyplot as plt
tips = sns.load_dataset("tips")
sns.kdeplot(
    data=tips,
    x="total_bill"
)
plt.title("KDE Plot")
plt.show()
```



```
sns.histplot(  
    data=tips,  
    x="total_bill",  
    kde=True  
)  
plt.title("Histogram with KDE")  
plt.show()  
  
#kde=True  
#adds a smooth density curve on top of the histogram
```



```
#figure size cintrols the width and height of the chart  
import matplotlib.pyplot as plt  
plt.figure(figsize=(9,3))  
plt.plot([1,2,3,4],[10,20,30,40])  
plt.title("Sales Trend")  
plt.show()
```



```
#subplots allow multiple charts in a single figure.Syntax:plt.subplot(rows,columns,position)  
#2 charts
```

```
#2 charts
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,4))
```

```
# First Plot
```

```
plt.subplot(1,2,2)
```

```
plt.plot([1,2,3], [10,20,30])
```

```
plt.title("Line Chart")
```

```
# Second Plot
```

```
plt.subplot(1,2,1)
```

```
plt.bar(['A','B','C'], [5,8,6])
```

```
plt.title("Bar Chart")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
#Example: Four Charts
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(10,8))
```

```
plt.subplot(2,2,1)
```

```
plt.plot([1,2,3], [10,20,30])
```

```
plt.title("Line")
```

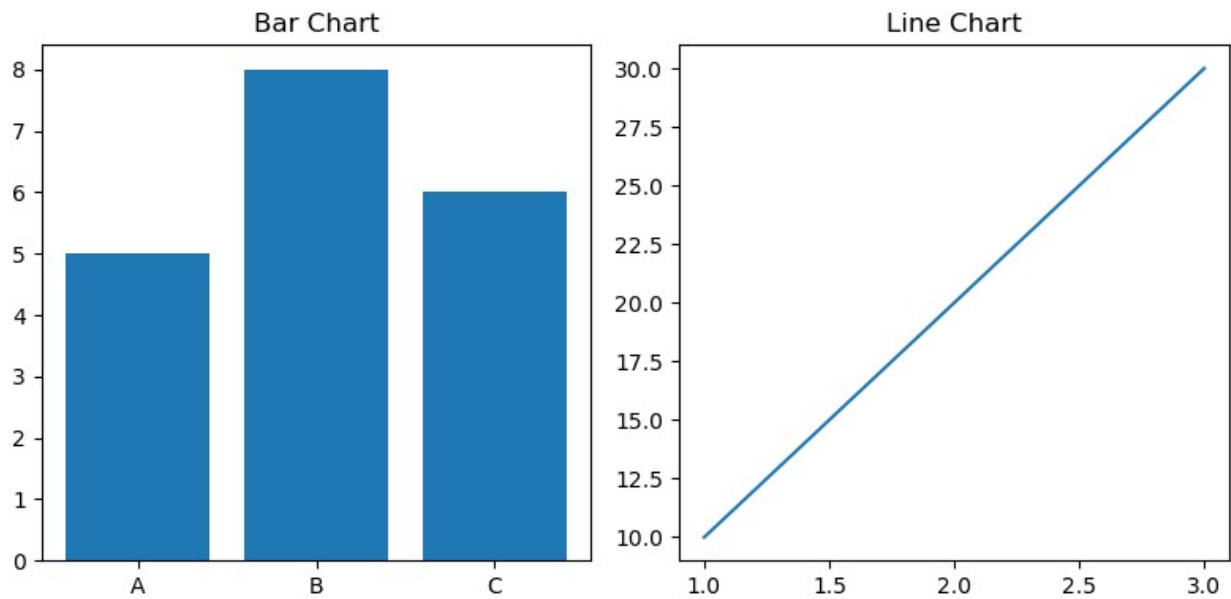
```
plt.subplot(2,2,2)
```

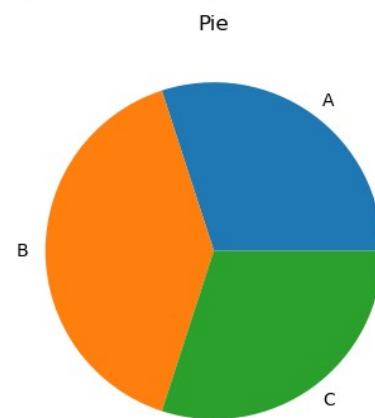
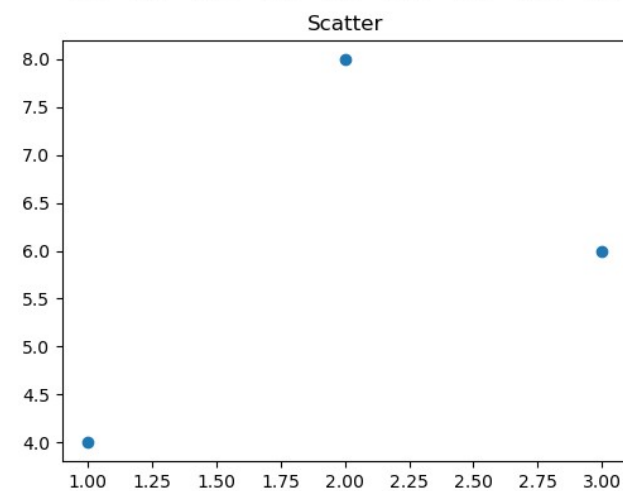
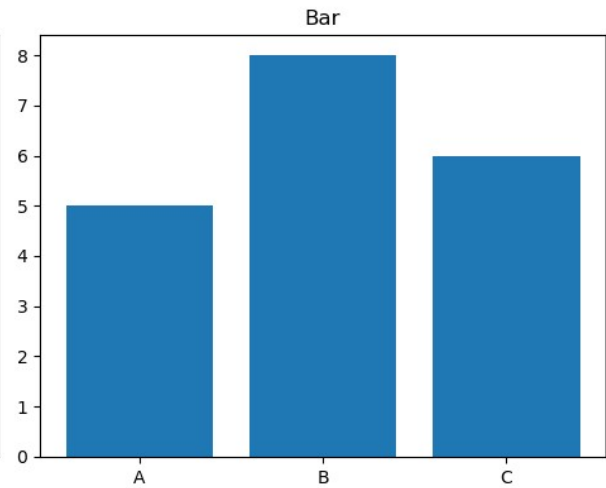
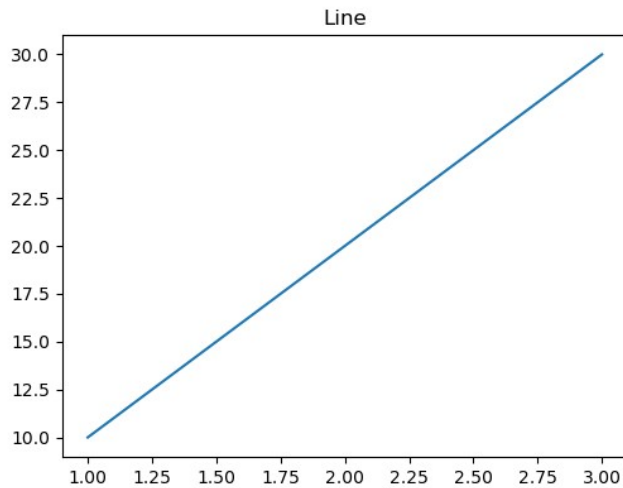
```
plt.bar(['A','B','C'], [5,8,6])
```

```
plt.title("Bar")
```

```
plt.subplot(2,2,3)
```

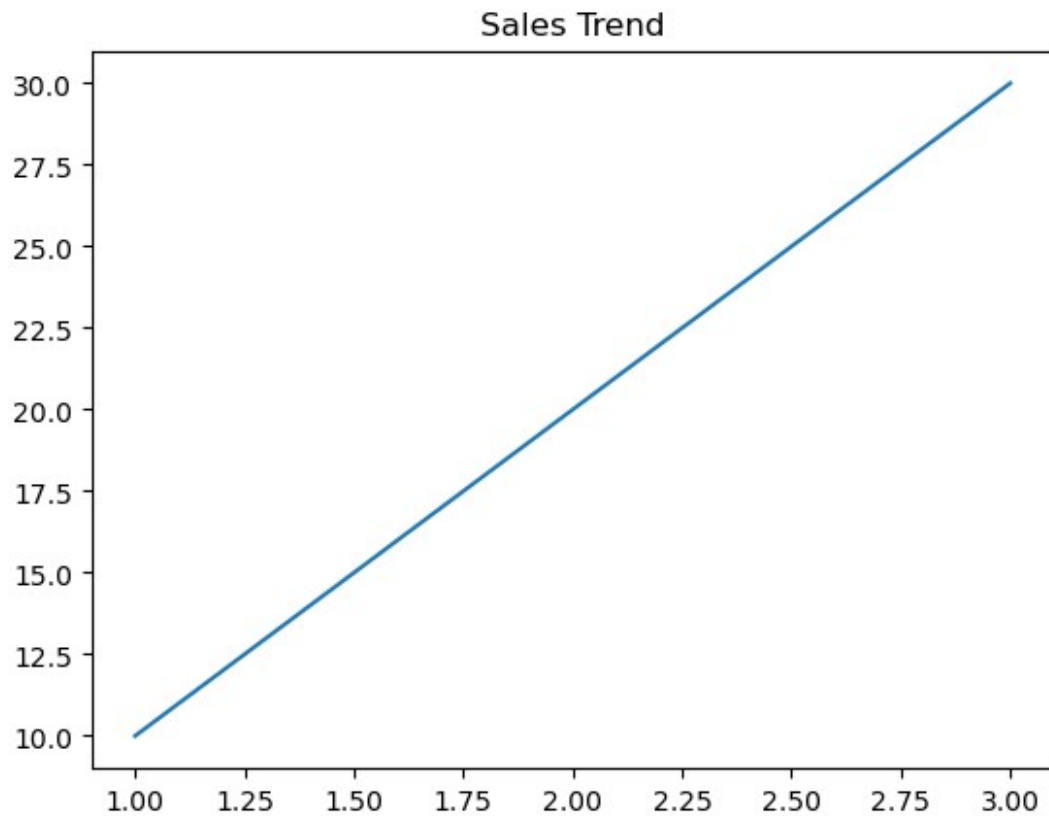
```
plt.scatter([1,2,3], [4,8,6])  
plt.title("Scatter")  
plt.subplot(2,2,4)  
plt.pie([30,40,30], labels=['A','B','C'])  
plt.title("Pie")  
plt.tight_layout()  
plt.show()
```





```
import matplotlib.pyplot as plt
plt.plot([1,2,3],[10,20,30])
plt.title("Sales Trend")
plt.savefig("Sales_chart.png")
plt.show()

#save with high quality
plt.savefig(
    "chart.png",dpi=300
)
```



<Figure size 640x480 with 0 Axes>